Windows Log Analysis
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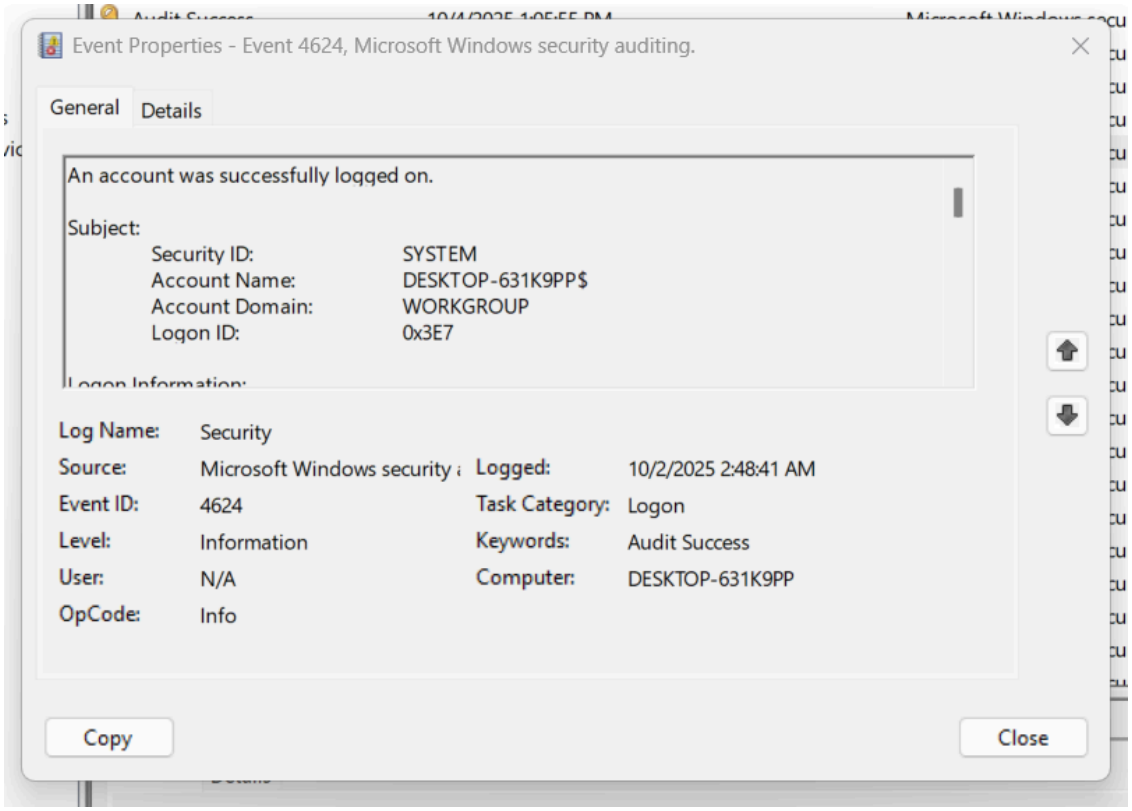
This lab was focused on analyzing real Windows Event Logs from my system using the Event Viewer application. Windows logs are one of the most valuable sources of forensic evidence because they record nearly everything that happens on a computer. Each log entry includes information like the event ID, time, source, user, and a message describing what occurred. By studying these logs, investigators can trace user actions, detect failed logins, identify system errors, and confirm when the system was started or shut down.

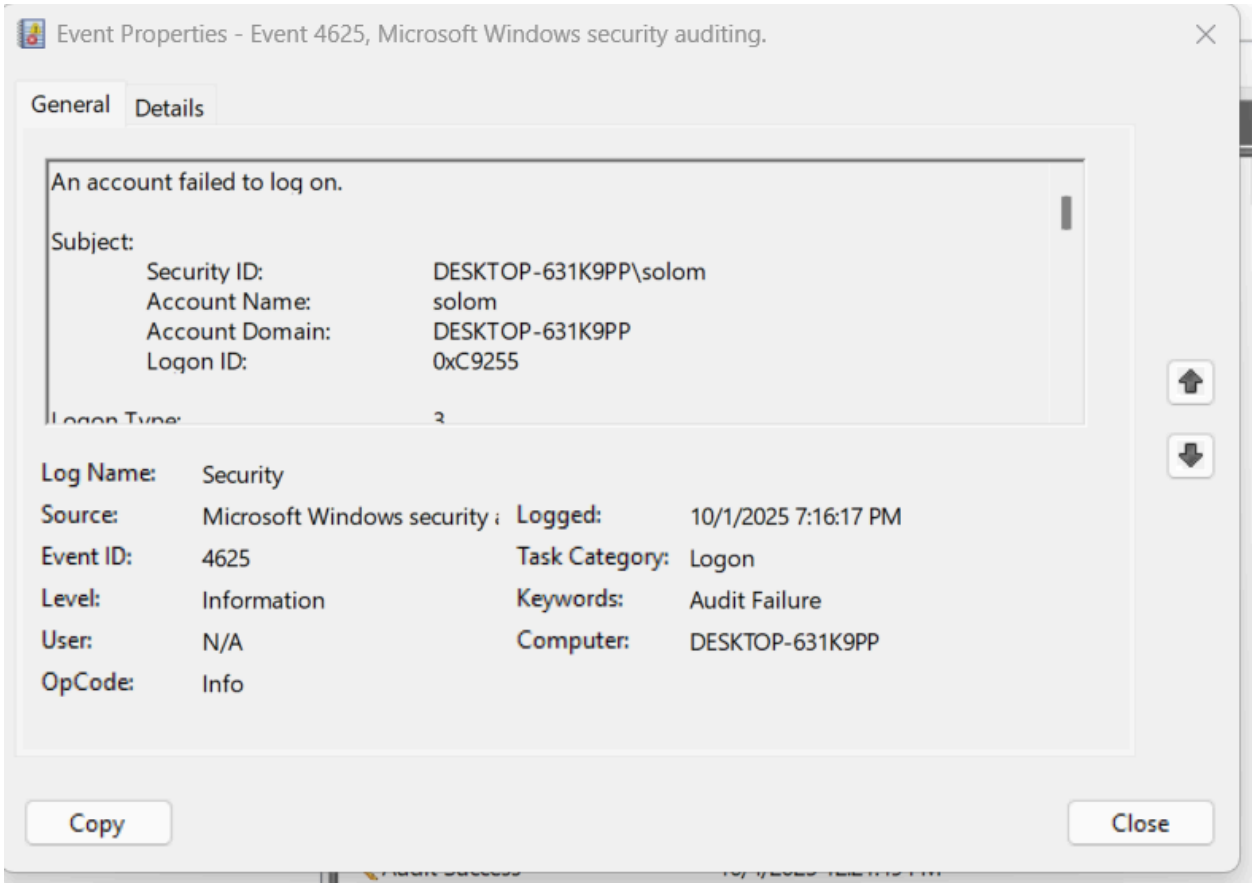
For this assignment I selected fifteen events from the Security, System, and Application logs on my computer. The goal was to understand what each event represents and how it could be useful in a forensic investigation. The screenshots that I captured from my Event Viewer are included throughout this report to show each specific event I analyzed.

Event 4624 – Successful Logon (Security Log)
This event records when a user successfully logs onto the system. It includes details like the account name, logon type, and computer name. From a forensic perspective, this event is critical because it confirms legitimate access and helps track who logged in and when.

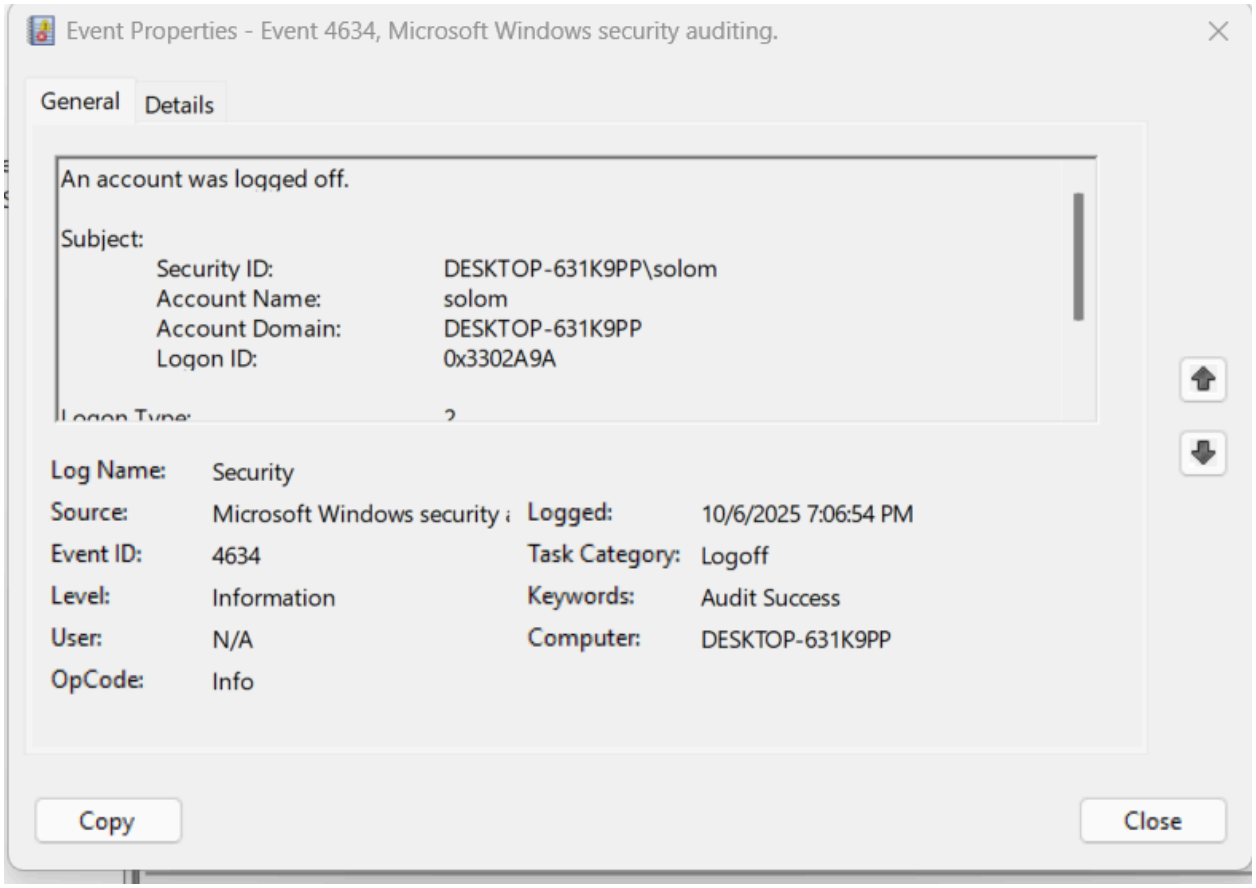


Event 4625 – Failed Logon Attempt (Security Log)

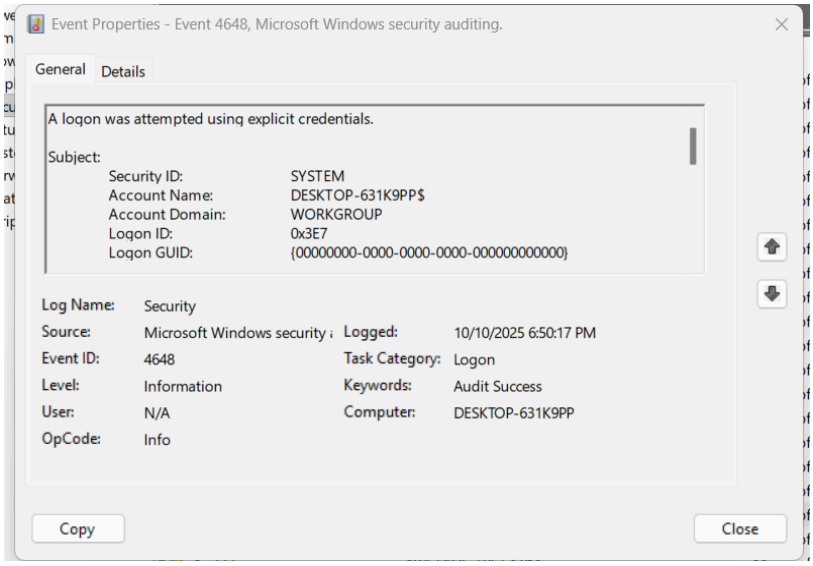
This event shows when a user fails to log in. It provides the account name and reason for failure. A series of failed 4625 events can point to password guessing or brute force attacks, making it very important when investigating possible break-in attempts.



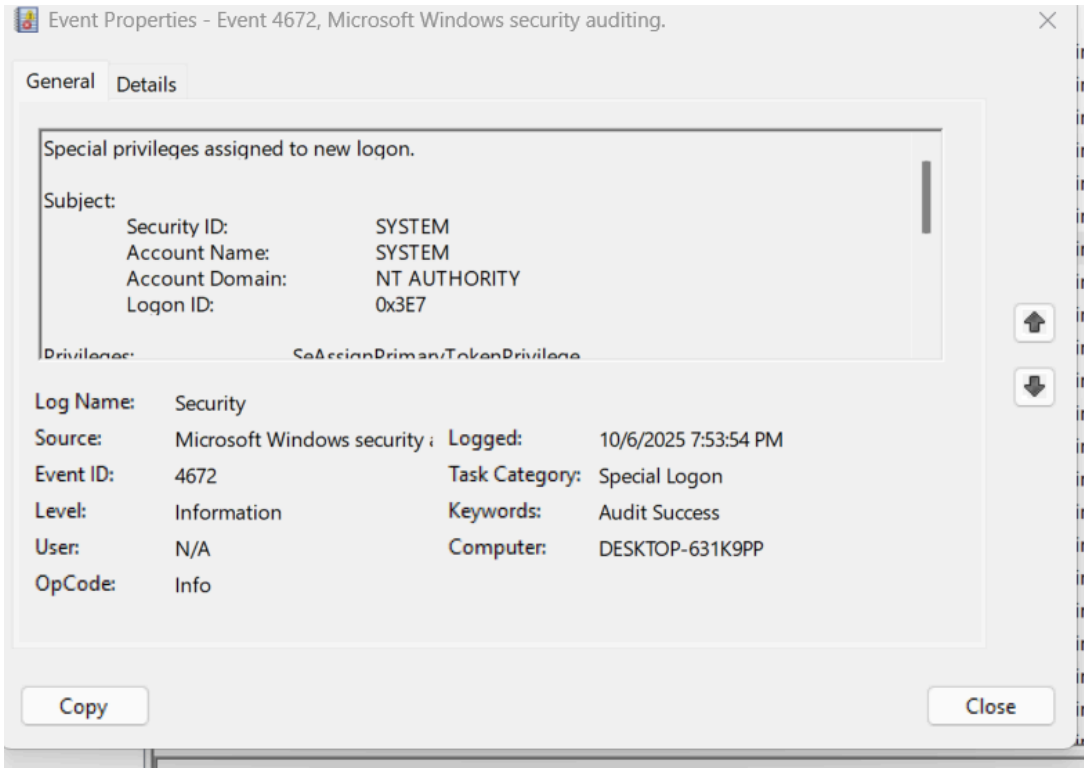
Event 4634 – Logoff Event (Security Log)
This event records when a user logs off from the computer. It can be used to match with the 4624 event to measure the duration of a session and to confirm when a user stopped using the system.



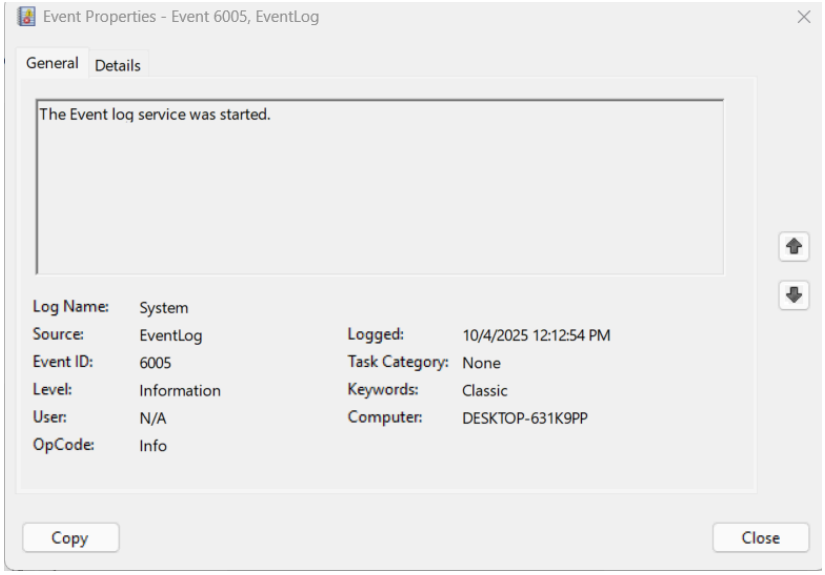
Event 4648 – Logon Using Explicit Credentials (Security Log)
This event appears when a user logs in with different credentials, for example using the “Run as Administrator” option. It can show privilege escalation or access from another account and is useful for identifying administrative actions.



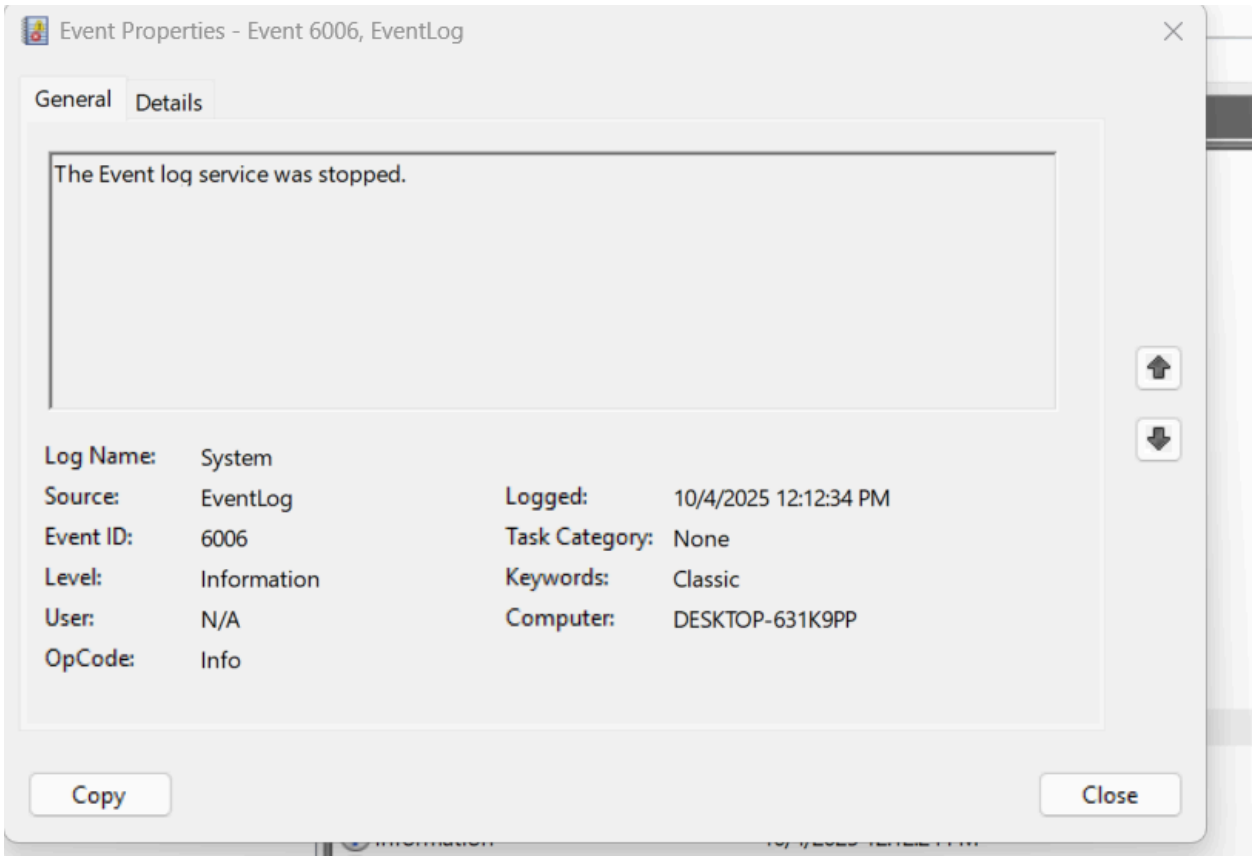
Event 4672 – Special Privileges Assigned (Security Log)
This event is recorded when a user is granted elevated or administrative privileges during login. It helps investigators see who was given special rights and at what time, which is key for detecting unauthorized admin activity.



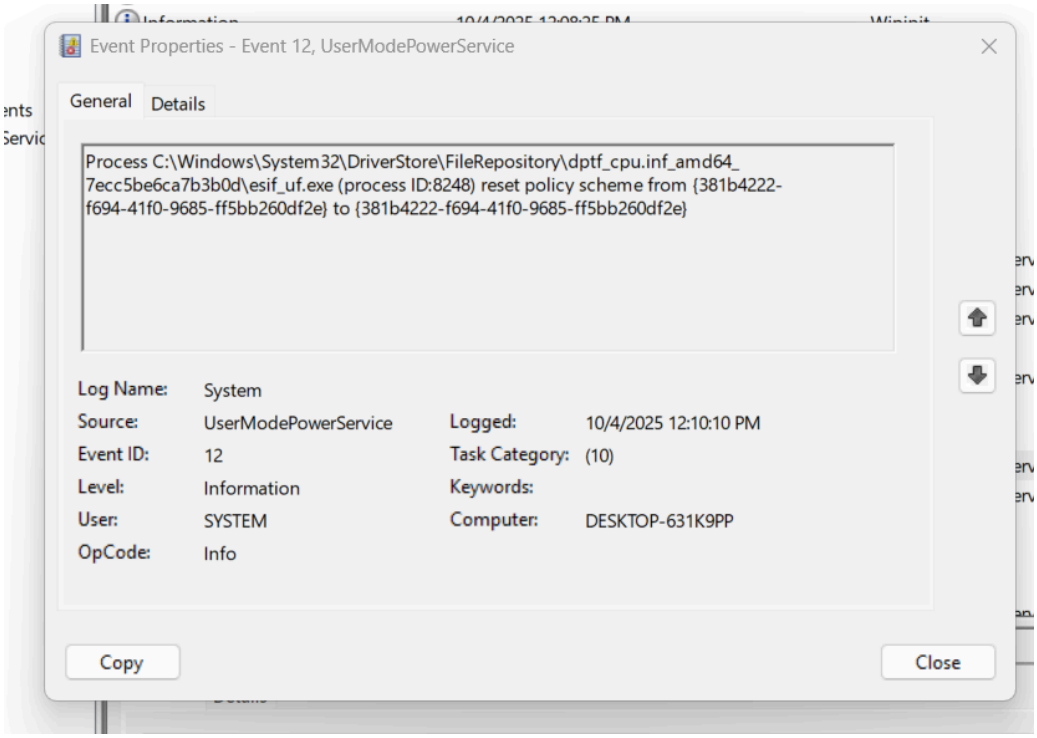
Event 6005 – Event Log Service Started (System Log)
This event shows that the Event Log service began running, usually right after the system boots up. It marks the start of a new logging session and helps determine when the computer was turned on.



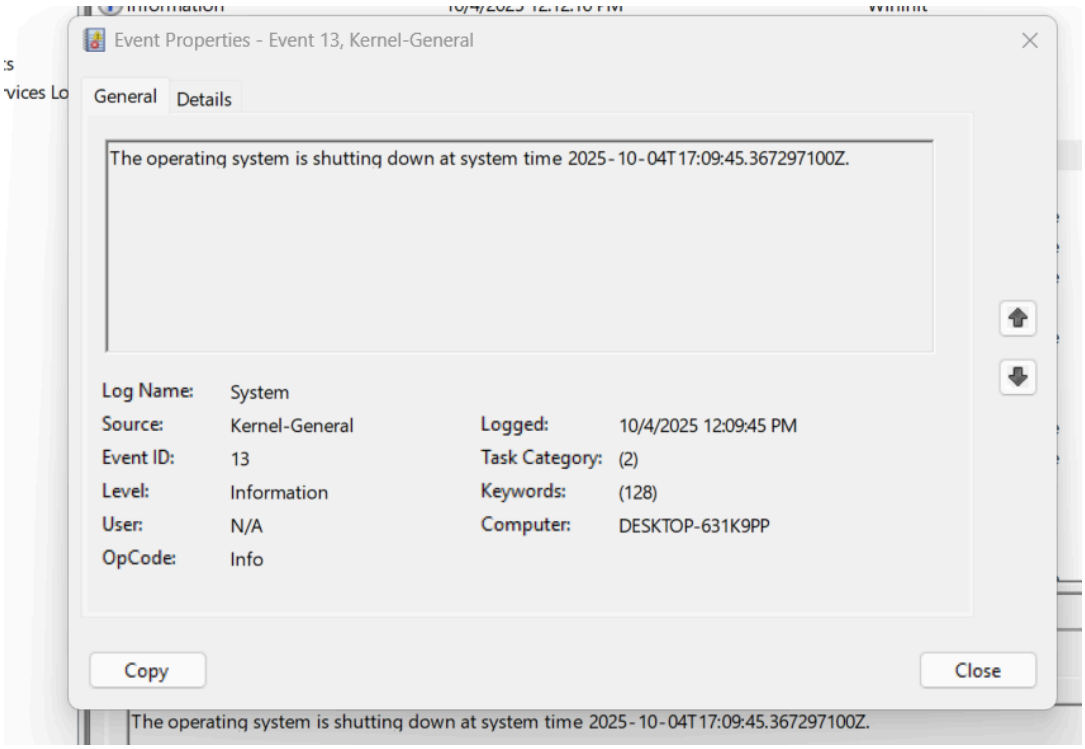
Event 6006 – Event Log Service Stopped (System Log)
This event means the Event Log service was stopped, which typically happens during shutdown. It can be matched with the 6005 event to confirm that the system was properly shut down and not powered off abruptly.



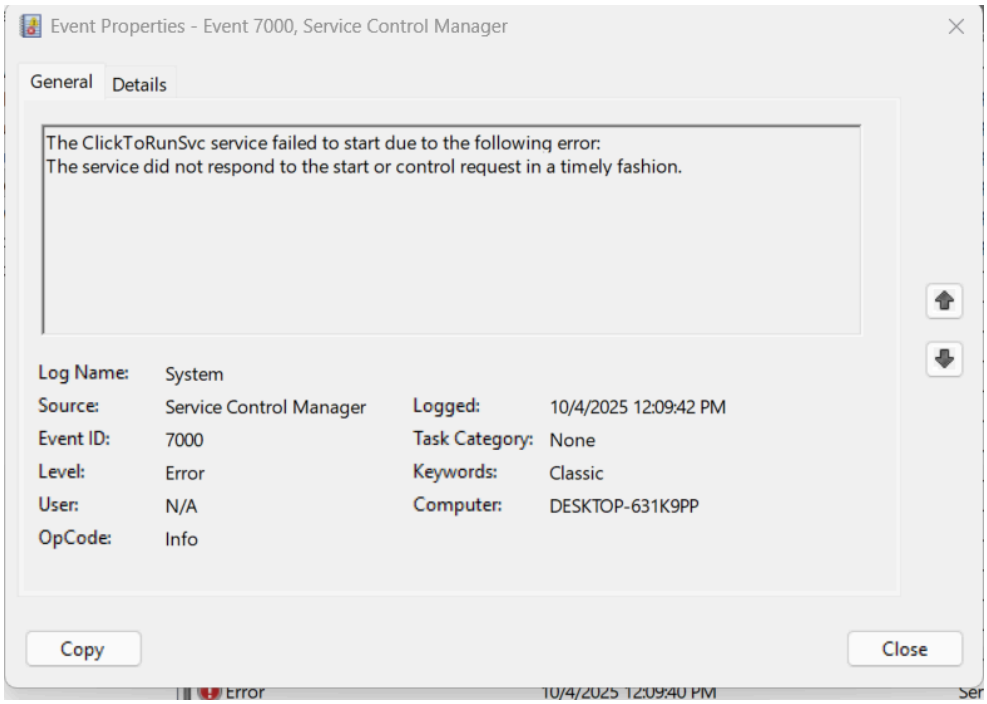
Event 12 – Operating System Start (System Log)
This event shows that the operating system has started its core processes. It helps verify the system boot time and is useful when building a complete timeline of user and system activity.



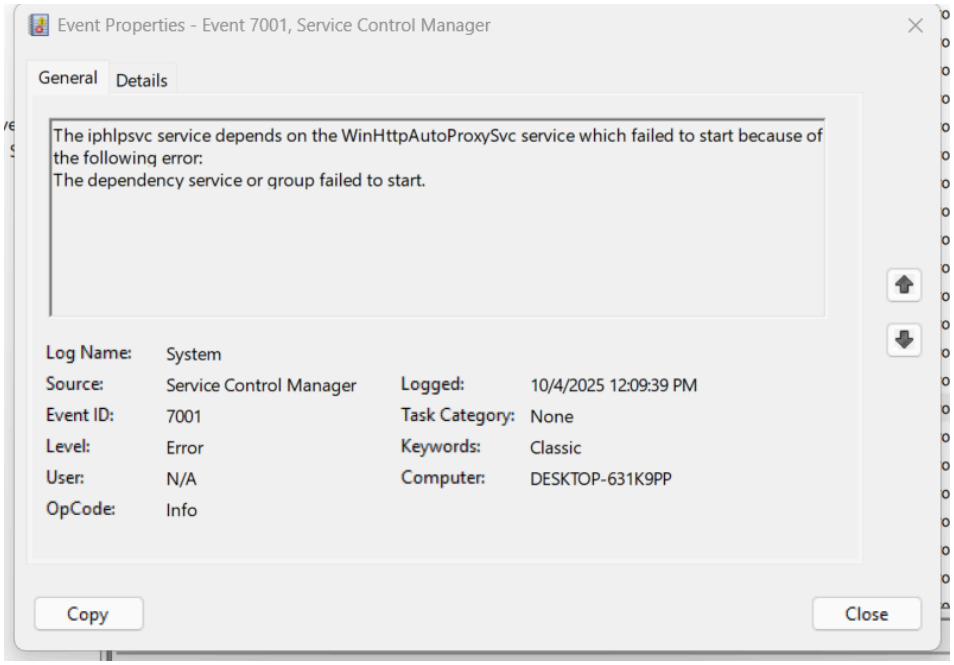
Event 13 – Operating System Shutdown (System Log)
This event indicates that the operating system has started the shutdown process. It provides the exact timestamp when the shutdown was initiated, which can be compared to the startup time to understand uptime and user activity duration.



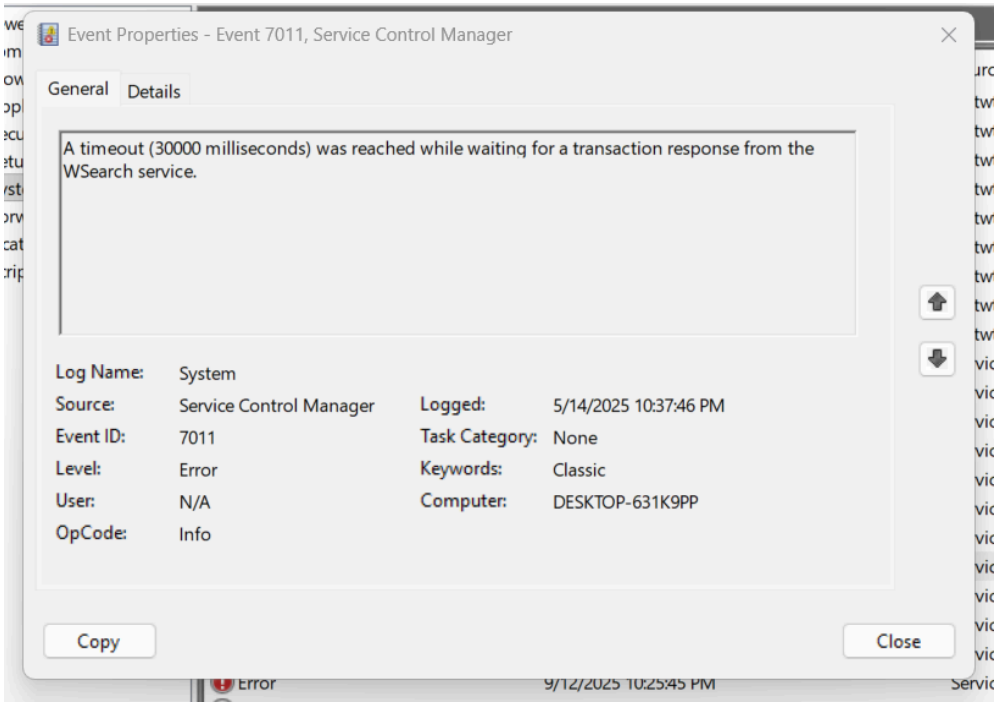
Event 7000 – Service Failed to Start (System Log)
This event appears when a Windows service cannot start. It lists the service name and reason for the failure. These entries can help identify malfunctioning or missing components and could also point to malicious services failing to launch.



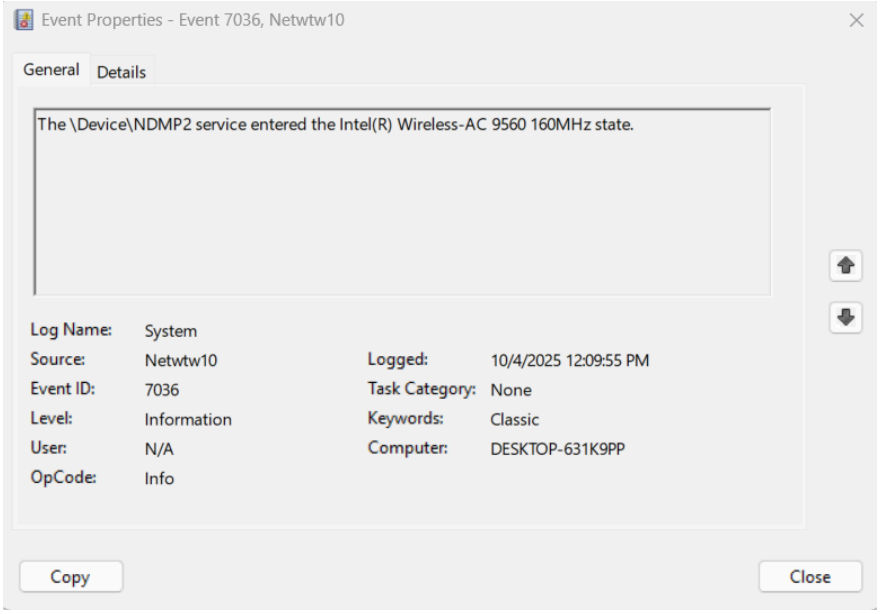
Event 7001 – Service Dependency Failed (System Log)
This event shows that a service could not start because it depends on another service that was unavailable. It helps forensic analysts track service dependencies and find root causes of startup issues.



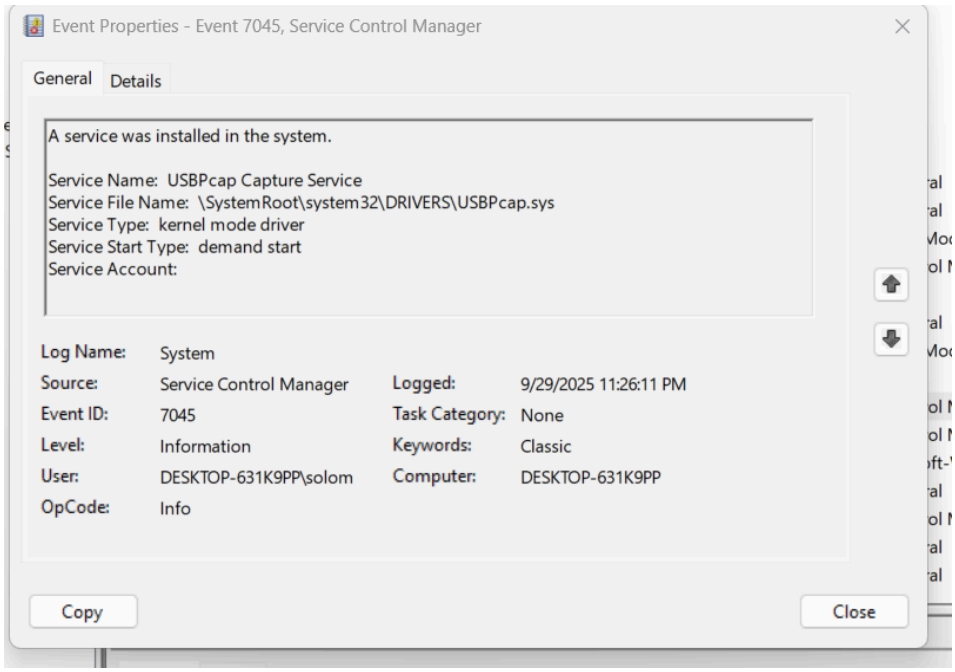
Event 7011 – Service Timeout (System Log)
This event means a service did not respond in the expected time frame. It can indicate system performance issues or stalled processes, and in some cases it can be a sign of interference from malware or corrupted system files.



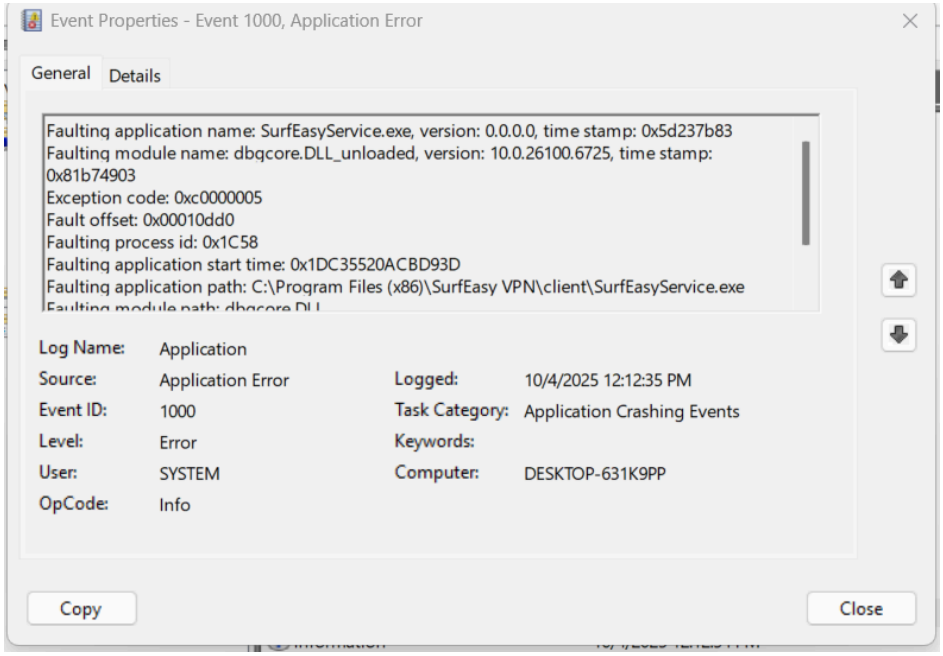
Event 7036 – Service Started or Stopped (System Log)
This event shows when a system service either starts or stops. It helps verify that important background services like networking or updates are functioning normally. It is also useful in forensic reviews to determine when key services were running.



Event 7045 – New Service Installed (System Log)
This event is recorded when a new service is installed on the system. It includes the service name, file path, and start type. In forensics this event is very important because attackers often create services to maintain persistence or execute malicious programs.



Event 1000 – Application Error (Application Log)
This event logs when an application crashes or stops working. It provides the application name, faulting module, and exception code. In forensic analysis, repeated application crashes can reveal instability, malware interference, or attempts to exploit certain software.



Conclusion

Reviewing and analyzing these fifteen Windows Event Logs helped me understand how detailed and valuable the logging system is for forensic work. Each event provides a snapshot of activity that, when pieced together, can tell a complete story of what happened on a computer. Security events show how and when users log in or attempt to log in. System events reveal information about startup, shutdown, and background services. Application events help identify software crashes or unusual behavior.

Overall, this lab taught me how to navigate Event Viewer, locate specific event IDs, and interpret the forensic significance of each one. These skills are essential in any digital investigation since log analysis can often be the key to uncovering suspicious activity or confirming legitimate actions.